A1 Week 2 Plan

Role 1 (Optimal Contract and General Carbon Theory)

- **Prove existence, uniqueness, and contract optimality.** Prove the interior optimum is unique via the recursive FOC structure — the distribution-free cutoff $F(K_F^\circ)=\phi^\circ$ pins K_F^* as a unique implicit function, then K_M^*, then K_R^* (a monotone-triangular / diagonal-dominance condition on the Hessian in primitives, not global concavity, which fails since Π is submodular in (K_R,K_F)); and prove $d^*(x)=\min\{(K_F-x)_+,K_M\}$ is the unique optimal dispatch via strict $R>c$.
- **Make the shift-ratio condition uniform along the homotopy.** Prove that support-wide monotonicity of $g(x)=F_{K_R}/f$ implies the realized endpoint condition holds uniformly along the whole C_M-homotopy, even as the window $(K_F^*-K_M^*, K_F^*)$ moves — this is the actual content Theorem 1's 'integrate downward' step needs, beyond the pointwise condition.
- **Prove the emission condition for general F (keystone).** Substitute the equilibrium map $K^*(C_R)$ into the exact inequality (C) to get the endogenous carbon condition, and prove a primitive sufficient condition (the endpoint shift-ratio condition plus a tail/skew bound) under which it holds at a finite interior optimum; exhibit a non-power-law F (e.g. truncated lognormal) satisfying it to prove the admissible class is non-vacuous.
- **Prove connectedness of the carbon-reduction set.** Prove $\{C_R : \Delta(C_R)\geq 0\}$ is bounded above (large-cost anchor) and non-empty, and that it is a single interval under an explicit monotone-renewable-shadow-value condition on F — general F can cross zero twice, so identify and prove the condition that rules out the second crossing.
- **Prove the sharp channel condition and tight bounds.** Establish the sharp necessary-and-sufficient condition for $K_R^*>K_R^\circ$: the realized two-point endpoint inequality $g(K_F^*)\geq g(K_F^*-K_M^*)$ — monotone g is sufficient but NOT necessary — and prove the two-sided bounds on the carbon saving Δ are tight by exhibiting the attaining distributions.

Role 2 (Solver and Numerical Methods)

- **Build and validate the general solver.** Implement the joint-capacity/dispatch solver for $d^*(x)$ and (K_R,K_F,K_M) under any residual-demand distribution, validated against the benchmarks to machine precision — including at least one non-power-law F with a semi-closed form, so the test set goes beyond the Week-1 uniform case.
- **Run the carbon-sign sweep at a finite optimum.** Sweep over Role 1's primitive condition on a synthetic, theory-tied distribution family and show numerically that carbon falls at a finite interior optimum — one principled map replacing the scattered Week-1 counts.
- **Derive the analytic Hessian and certify negative-definiteness.** Derive and implement the full analytic Hessian and comparative-statics cofactors (replacing the knife-edge-noisy finite differences), and on that analytic Hessian certify negative-definiteness along the interior path for Role 1's class.

- **Map the boundary of the regularity region.** Exhibit distributions in the admitted class where the Hessian fails to be negative definite, delimiting Condition 2 sharply and complementing Role 1's uniqueness proof — a numerical map of where regularity holds.

Role 3 (Comparative Statics and Robustness)

- **Sign the price and adoption-margin comparative statics.** Via the Hessian-cofactor identities, under the explicit interior-regime hypotheses (R inside the Assumption-1 bounds), sign ∂(carbon effect)/∂R, ∂/∂β, ∂/∂p, and ∂/∂C_M (the adoption margin), and prove mining raises $K_R^* > K_R^\circ$ under the endpoint shift-ratio condition.
- **Prove the cross-distribution robust certificate.** Along a specified one-parameter deformation within the shape class U (a mean-preserving-spread / likelihood-ratio path), sign the capacity covariation dK*/dt so the least-favorable result holds across the path — making $C_R \le C_R^\dagger \Rightarrow \Delta \ge 0$ a genuine robust certificate — or prove the obstruction.
- **Construct the reversed-channel counterexample.** Construct a realized F violating the endpoint shift-ratio condition ($g(K_F^*) < g(K_F^* - K_M^*)$ at the realized thresholds) for which $K_R^* < K_R^\circ$, and solve it numerically — the sharpness witness Week 1 could not produce, certifying necessity of the endpoint condition (Role 1).
- **Prove the limit-interchange rigorously.** Replace the expected-fossil-to-zero a.s.-corner step with a dominated-convergence / finite-capacity envelope argument, so the threshold proof holds without the limit shortcut.

Role 4 (Mechanism Design and Regimes)

- **Prove a single-price implements the optimum.** Prove the non-trivial instrument result: a single per-MWh price R (not a state-contingent transfer) uniquely implements the dispatch d* and pins the capacity vector, and characterize exactly when the price instrument is rich enough to do so — the regulator's instrument-design problem.
- **Prove the unified regime theorem.** Compose the C_M coverage axis (no adoption / small-FCL / interior / full coverage) and the C_R boundedness axis into one theorem, and pin down the small-FCL/large-FCL crossover precisely as the parameter-space locus where $K_M^* = K_R^\circ$ (left open in Week 1).
- **Prove the necessity of Assumption 1.** Upgrade the four parameter conditions from pressure-test to proof: prove each is necessary for the interior optimum or the sign it guards, and characterize the regime obtained when relaxed (full-coverage corner, no adoption, or perverse curtailment).
- **Characterize the second-best contract under private information.** Introduce genuine private information the principal cannot infer from observed demand — the agent privately knows the renewable factor S while only the demand D is observed, so residual demand is the agent's private signal — and derive the second-best screening contract and how its information rent shifts the emission outcome versus the full-information benchmark.